

Floating-Rate Notes

Floating-rate notes (FRNs) may appear to sit oddly in a book of fixed income instruments. They are bonds where the coupon is not fixed but is reset, usually at the start of each interest period. However, as will be discussed below, under some circumstances this feature stabilises the value of the bonds compared to other securities. The term variable-rate note is used almost interchangeably with the term FRN but has slight connotation of a coupon structure that is linked to something else than benchmark interest rates.

The cashflow structure of an FRN is shown in Figure 17.1. The specific features of an FRN are the choices for the reference index used to calculate the coupon rate, the coupon frequency, and the timing of the resets. In the majority of FRNs, coupons reset to a 'natural' index at the beginning of each coupon period. However, more complex FRNs exist and the full spectrum of FRNs ends at highly structured products. In essence, if one thinks of a bond coupon as a function of some inputs:

$$C = f(\text{something}) \quad (17.1)$$

then the standard bullet bonds discussed so far are the most simple manifestation of this equation, namely:

$$f(\text{something}) = \text{const} \quad (17.2)$$

Floating-rate notes are bonds where f takes a more interesting form. Although the vast bulk of FRNs has a very natural structure for f , some can be very complex.

The typical investor in a floating rate bond is the treasury of a deposit-taking institution, for instance a retail bank. Banks compete against other banks and other products for deposit money. The deposit rate each bank offers is therefore a compromise between wishing to pay a low rate (to save costs) and a high rate (to attract money). At the same time, deposit funding competes with other funding sources that may be available to the bank. As a result, deposit interest rates tend to follow interbank market funding interest rates, albeit usually with a spread that reflects the higher cost of deposits. This extra cost is linked to the need to advertise, retain customer service staff, and maintain buildings that impress the saving public. Although deposit interest rates fluctuate, the deposits themselves tend to remain at a bank for longer periods. Customers do not frequently move deposit money between different banks, and for this reasons deposits are said to be 'sticky'.

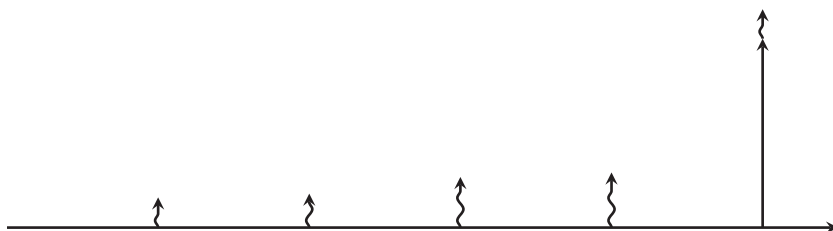


FIGURE 17.1 Simple floating rate note. The bond repays its par amount at the maturity date. The bond has regular accrual periods but coupons are reset at the beginning of each coupon period in accordance with some pre-defined index.

As a result of this economic setting, the liability side of a bank tends to be linked to short-term rates in the interbank market. Banks therefore are natural investors in assets that also pay the variable interbank rates, or use derivatives to transform the cashflow structure of other instruments into a variable-rate structure.

By the same logic, some retail investors purchase FRNs as an alternative to holding term deposits at a bank that roll over periodically. The advantage of doing so can be a higher coupon, but also that a FRN can be sold at any time while a term deposit can usually only be terminated at maturity without incurring extra penalties. Also, only banks can offer term deposits while FRN can be issued by other entities.

17.1 COUPON RESET MECHANICS

FRNs usually reset the coupon at the beginning of the coupon period. There can be a lag between the observation of the benchmark and the reset date. For instance, because money market deposits settle $T + 2$ in most markets, the coupon used for a coupon period starting at T will usually be the reference rate taken at $T - 2$. Some FRNs, particularly more structured ones, have longer lags which sometimes reach 20 business days, i.e., almost a month.

It is possible to fix the coupon later than at the start of the accrual period. Such in-arrears structures can delay the coupon fixing to as late as the payment date. It is meaningless to calculate an accrued interest when the accrual rate is not fixed and in-arrears structures therefore trade on dirty, rather than clean price.

Coupons only rarely set to the benchmark itself, and instead a spread is set at issuance, and then applied at every reset date. This spread is called the quoted margin, called here μ . The coupon on a standard FRN payable over the period starting at t_i would therefore be:

$$C_i = R(t_i - 2) + \mu \quad (17.3)$$

where as usual all rates are quoted in annualised terms even though FRN coupon periods are frequently shorter than one year. Note that other formulas are possible.

A so-called inverse floater for instance uses a formula like:

$$C_i = \max(0, \mu - R(t_i - 2)) \quad (17.4)$$

so the coupon moves in the opposite direction of the benchmark.

Most jurisdictions do not allow negative coupons which would be payment obligations from the investor to the issuer. In addition, many CSDs are unable to process such payments. Implicitly, therefore, Equation (17.3) is to be read as:

$$C_i = \max(0, R(t_i - 2) + \mu) \quad (17.5)$$

This implicit coupon floor is now standard in bond documentation because there have been frequent instances where the simple formula Equation (17.3) would have led to negative resets. Again, nothing prevents issuers from setting the floor to any other non-negative value and some FRNs have floors above zero. In the same vein, issuers can furnish coupons with a cap.

17.2 LIBOR AND OIS-LINKED NOTES

The ‘natural’ coupon of a floating rate note is the interbank benchmark rate applying to the period between two coupon resets, as pertaining at the start of the accrual period, and the coupon formula Equation (17.3). A quarterly FRN would therefore have coupon resets linked to the 3M Libor rate applicable in that currency.

If that is the case, then the present value of the FRN at the time of its last coupon reset (i.e., one coupon period before maturity) would be:

$$P = \frac{100 + C * DCF}{1 + rDCF} \quad (17.6)$$

where r is the appropriate discount rate applicable to liabilities of the FRN issuer. This discount rate should be the market rate plus an appropriate spread. A quick comparison with Equation (17.3) shows that if that spread happens to be μ , then $P = 100$ in this equation. At the beginning of the last coupon period, the FRN is therefore worth exactly par if the quoted margin is equal to the credit spread of the issuer. One can now consider the beginning of second-to-last coupon period. At the end of that period, the bond will be worth par (as has just been shown), and an additional coupon will be paid on the end of that period. Equation (17.6) wherefore applies again, only that in this instance the 100 in the numerator refers to the value of the bond, not its redemption amount. It follows that also at the start of the second-to-last coupon period the bond has a value of par. This induction can be continued all the way to the first coupon period. The result is that the value of an FRN is equal to par at the time of each coupon reset if the quoted margin is the correct risk premium for the issuer.

Between coupon resets the FRN value can fluctuate because the market discount rate applicable between the settlement date of any trade, and the market rate when the coupon was fixed can differ. For instance, if the central bank makes a surprise rate cut

after a coupon has been fixed, the next coupon payment will reference the older, higher central bank rate, and so the FRN price will increase to reflect the lower discount rate. However, the effect of this mismatch between the coupon fixing and the market rate will end with the next coupon payment. In total therefore, FRNs have a very stable market price as long as the risk premium of the issuer remains constant.

As mentioned in the section on Libor, Section 13.6, unsecured term benchmark rates are being phased out globally because regulators have taken the view that benchmarks based on an economically irrelevant subset of transactions are open to abuse. FRN issuers have therefore switched to coupons that follow overnight benchmark rates. In principle, the coupon payment (not the annualised coupon rate) should amount to the compounded interest on a money market account:

$$c(t-1) = \prod_{t=t_0}^{t_1} [1 + (r(t-2) + \mu)DCF(t, t+1)] - 1 \quad (17.7)$$

where $r(t-2)$ is the overnight rate observed for each day t applicable to the relevant overnight rate (note that money markets settle T+2) within the coupon period, and $DCF(t, t+1)$ is the applicable daycount fraction¹. This formula, however, has a structural disadvantage and therefore is not how interest is calculated on actual bonds. The total interest amount is only known at the end of the coupon period and so it would be difficult to define accrued interest during the accrual period. Inconveniently for the issuer, the spread μ would also be compounded, and so be not directly comparable to the quoted margin of a standard Libor FRN.

Instead, Equation (17.7) is used to construct an intermediate term rate which is the annualised compound interest between the last coupon payment and any settlement date within the coupon period, including the next payment date:

$$\rho(t-1) = \frac{\prod_{t=t_0}^d [1 + (r(t-l))DCF(t, t+1)] - 1}{DCF(t_0, d)} \quad (17.8)$$

$\rho(t)$ is then used in the same way as $R(t)$ in Equation (17.3) to calculate the coupon fixing. In this way, the quoted margin μ is not compounded, and the formula can be used to calculate interim accrued interest. This accrued interest is in effect based on the best possible estimate of the eventual coupon payment. Somewhat unintuitively, the standard observation lag l between the overnight rate and its application appears to have been selected in the European market to be five business days rather than two².

¹The daycount fraction is not simply 1/360 for most currencies and 1/365 for the Commonwealth markets because there may be a weekend or public holiday between t and $t+1$.

²This convention enables a longer-than-standard T+2 settlement while retaining defined accrued interest.

17.3 DISCOUNT MARGIN

Floating-rate notes do not have predictable cashflows and therefore they do not have yields. One approach to compare expected returns of FRNs is the so-called fixed-equivalent yield where future cash flows are projected, either by simply assuming constant coupons, or that future reference index levels can be predicted from some form of forward curve. In the latter case, the actual coupon formula, e.g. Equation (17.3), can be used directly. Once cashflow projections are obtained, a yield can be calculated from this series of irregular cashflows in the same way as for any other bond.

The more common market convention, however, is to calculate a spread known as the discount margin. Conceptually, the discount margin μ' is the quoted margin that would make the price of a given FRN equal to par. That suggests that one should calculate it based on a discount factor term structure as:

$$\mu' = \mu - \frac{P - 100}{\sum_i DCF(t_{i-1} - t_i) Df(t_i)} \quad (17.9)$$

where the denominator of this fraction acts to distribute the difference between the market price and par across each coupon period, bearing in mind that like the quoted margin, the discount margin is quoted in annualised terms and so needs to be de-annualised for the daycount fraction of each coupon period. As presented here, the discount margin is more closely related to the notion of an asset swap spread (cf. Section 22.3.1 on page 222).

Equation (17.9) is straightforward if one has a discount factor curve $Df(t)$ to hand but amounts to relying on external instruments to build that curve. The market convention is to calculate the discount margin from the bond alone. The approach resembles that of the yield calculation, with the added assumption that the benchmark resets in the future will be identical to the current rate. Specifically, the dirty price of the bond P is expressed as:

$$P = \frac{(R(t_l) + \mu) DCF(t, t_1)}{1 + DCF(t, t_1)(R^*(t) + \mu')} + \sum_{i=2}^n Df(j) DCF(t, t_i) (R(t) + \mu) + 100 Df(n) \quad (17.10)$$

where the discount factors $Df(k)$ are given by the recursive relationships:

$$Df(k) = \frac{Df(k-1)}{1 + DCF(t_{k-1}, t_k)(R(t) + \mu')} \quad (17.11)$$

and:

$$Df(1) = \frac{1}{1 + DCF(t, t_1)(R^*(t) + \mu')} \quad (17.12)$$

Here $R(t_l)$ stands for the last fixing of the reference index while $R^*(t)$ is the money market rate between the settlement date and the next coupon date (the so-called stub rate). The discount factors $Df(k)$ amount to the compounded discounted value of a deposit rolling at the frequency of the FRN with an interest rate equal to the current fixing of

the benchmark rate plus the fixed spread μ' which is what this nonlinear equation needs to be solved for. The three terms in Equation (17.10) correspond to the discounted values of the next coupon, all subsequent coupons, and the principal, in that order. This equation needs to be solved for μ' using a non-linear solver, similar to calculating a compound yield.

In essence, the difference between Equations (17.9) and (17.10) is that the former would lead to a different result if the forward curve of the reference rate moved, while the latter would not. When the yield curve steepens, the discount factors $Df(t_i)$ in Equation (17.9) decline, so the asset swap spread for a given price increases. The definition Equation (17.10) does not depend on the current yield curve at all (only on the two rates $R^*(t)$ and $R(t)$), so the discount margin is unaffected by curve changes. That means, just as for yields, that it is methodologically incorrect to compare discount margins across bonds of different maturities.

When the FRN is linked to overnight rates like €STR, Equation (17.10) is no longer appropriate because the term rates $R^*(t)$ and $R(t)$ do not exist, and the next coupon $R(t_i) + \mu$ is unknown. The sensible solution is to use the same $\rho(t)$ from Equation (17.8) instead of these term rates.

17.4 CMS AND CMT FLOATERS

Several entities, including the French, Italian and Japanese governments, have issued variable-rate bonds where the coupons are set with reference to a long-term rate. In the French and Japanese cases, that rate was a 10Y government yield, while Italy issued eurobonds with coupons linked to various swap rates, usually the 10Y swap rate. Bonds linked to benchmark yields in the government bond and swap markets are known as constant-maturity treasury (CMT) and constant-maturity swap (CMS) floaters. Private issuers also issue securities with more complex CMS structures, such as coupons linked to the spread between 2Y and 10Y CMS rates³.

The first question about such bonds is how the relevant rates can be observed, given that bond and swap markets are over-the-counter markets where prices are negotiated in private. For CMT rates, the French market has a well-known benchmark known as the TEC10 which is calculated from price quotations provided by the primary dealers in French bonds. In Japan, the yields seen in the most recent government bond auction, given the large volume transacted in such auctions, give a reliable pricing. For CMS rates, standard fixings exist in the same way as Libor fixings⁴. Such fixing rates are now subject to the same governance rules as other benchmark rates.

Because the fixing rate for the coupon does not coincide with the discount rate applicable to the coupon period, Equation (17.6) does not have the effect of keeping bond prices close to par. Instead, CMT and CMS structures have complex exposures

³This particular structure would be a 2-10 CMS steepener.

⁴At the time of writing, the Intercontinental Exchange (ICE) provides swap rate fixings for various currencies. These fixings were in the past administered by the International Swap Dealers Association and are therefore still sometimes colloquially called ISDAfix.

to long-term interest rates given that increasing long-term rates have both the effect of higher forward rates (and thereby expected coupon resets) and lower discount factors. The two effects partly counteract each other in their impact on bond prices so that the convexity adjustment in their pricing is substantial.

CMS structures are generally easier to hedge than CMT structures because paying or receiving swaps has a lower balance sheet cost than holding long or short positions in bonds with the attendant repo operations. As a result, CMT portfolios are usually hedged with CMS structures, exposing the hedge to basis risk. In this case, the risk is a change in the CMS-CMT spread, which is in essence the swap spread of the government bond curve.